Action and the Calculus of Variations

Why mechanics compares whole paths

From a small change of a curve to the Euler–Lagrange equation

How to read this tutorial

Pause at each boxed result and ask what it says geometrically or physically. The derivation is part of the explanation, not a hurdle to skip.

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1 The new kind of question

Ordinary calculus asks how a number changes when we change another number. If

$$f(x) = x^2,$$

then $f'(x)$ measures the response to a small change $x \mapsto x + \varepsilon$.

The calculus of variations asks a similar question about an entire function. A path

$$q : [t_0, t_1] \longrightarrow \mathbb{R}$$

is the object being changed. A *functional* assigns one number to the whole path. The central example in mechanics is the action

$$\boxed{S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t)\ \mathrm{d}t.}$$

The square brackets are a useful reminder: S consumes a function $q(t)$, not merely its value at one instant.

A landscape whose points are curves

Imagine that every admissible path is one point in a vast "space of paths". The action gives a height to each point. A physical path is stationary when every allowed infinitesimal direction through this space produces no first-order change in height.

2 Why an integral over time?

The Lagrangian $L(q, \dot{q}, t)$ measures something locally, at one instant. The action adds that local contribution along the entire history. For an ordinary conservative particle,

$$L = T - V.$$

The action therefore balances motion and potential along a path rather than choosing the smallest energy at every instant.

Stationary does not always mean smallest

The physical path makes the first variation vanish. It may give a local minimum, a local maximum, or a saddle value of the action. "Principle of stationary action" is more accurate than "principle of least action".

3 How to vary a path

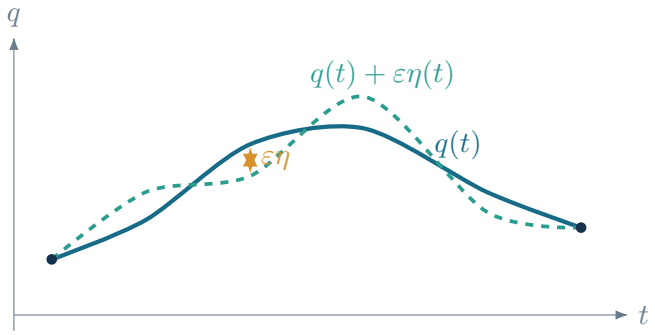
Choose a reference path $q(t)$ and a smooth shape $\eta(t)$. Define a one-parameter family

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t).$$

The small number ε controls the size of the change; η controls its shape.

If the endpoints are fixed, every neighbouring path must begin and end at the same places, so

$$\boxed{\eta(t_0) = \eta(t_1) = 0.}$$



Because differentiation is linear,

$$\dot{q}_\varepsilon = \dot{q} + \varepsilon\dot{\eta}.$$

Changing the path therefore changes both the position argument and the velocity argument of L.

4 The first variation

Insert the varied path into the action:

$$S[q_\varepsilon] = \int_{t_0}^{t_1} L(q + \varepsilon\eta, \dot{q} + \varepsilon\dot{\eta}, t)\ \mathrm{d}t.$$

Now $S[q_\varepsilon]$ is an ordinary function of the number ε. A stationary path satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0} = 0.$$

Using the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[q_\varepsilon]\bigg|_{\varepsilon=0} = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta} \right) \mathrm{d}t.$$

This expression is called the *first variation* and is written δS.

5 The decisive integration by parts

The second term contains $\dot{\eta}$, but we want the arbitrary function itself. Integrate by parts:

$$\int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}}\dot{\eta}\ \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}}\eta \right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \eta\ \mathrm{d}t.$$

The boundary term vanishes because $\eta(t_0) = \eta(t_1) = 0$. Hence

$$\delta S = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta(t)\ \mathrm{d}t.$$

The function η may be chosen freely in the interior. The only way this integral can vanish for every such choice is for its coefficient to vanish point by point.

Euler–Lagrange equation

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.}$$

One integration by parts converts a statement about every nearby path into a differential equation for the stationary path.

6 Example: the free particle

Let

$$L = \frac{1}{2} m \dot{q}^2.$$

Then

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}, \qquad \frac{\partial L}{\partial q} = 0.$$

The Euler–Lagrange equation gives

$$m\ddot{q} = 0.$$

Thus the stationary path has constant velocity:

$$q(t) = At + B.$$

Why a straight world-line?

With no potential, unnecessary changes of velocity only increase the kinetic contribution. Between fixed endpoints and fixed times, the stationary path is the line of uniform motion.

7 Example: the harmonic oscillator

For a mass on a spring,

$$L = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} k q^2.$$

The two required derivatives are

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}, \qquad \frac{\partial L}{\partial q} = -kq.$$

Therefore

$$\boxed{m\ddot{q} + kq = 0.}$$

The familiar restoring-force equation has emerged from stationarity of a whole-path quantity.

8 Many coordinates

If the configuration is described by $q_1, \ldots, q_n$, vary each independently:

$$q_i \mapsto q_i + \varepsilon \eta_i.$$

The result is one equation per coordinate:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0, \qquad i = 1, \ldots, n.}$$

This is why a single scalar function L can encode the dynamics of a many-coordinate system.

9 What if the endpoint is free?

When an endpoint value is not fixed, the corresponding value of η need not vanish. The boundary term

$$\left[\frac{\partial L}{\partial \dot{q}}\eta\right]_{t_0}^{t_1}$$

must then be handled explicitly. For example, if $q(t_1)$ is free while t_1 is fixed, stationarity requires

$$\left.\frac{\partial L}{\partial \dot{q}}\right|_{t_1} = 0.$$

Such conditions are called *natural boundary conditions*. They are not extra guesses; they are the part of the variational calculation that fixed endpoints previously hid.

10 Adding a total derivative

Suppose

$$\widetilde{L} = L + \frac{\mathrm{d}}{\mathrm{d}t}F(q,t).$$

Then

$$\widetilde{S} = S + F(q(t_1),t_1) - F(q(t_0),t_0).$$

For fixed endpoints the added quantity is fixed, so it does not change the stationary paths. Different-looking Lagrangians can therefore encode the same equations of motion.

11 From particles to fields

For a field $\phi(x,t)$, the action becomes an integral over space and time:

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu\phi, x)\ \mathrm{d}^{d+1}x.$$

Varying the field and integrating by parts in every coordinate gives

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0.}$$

The logical structure is unchanged: vary the whole object, move derivatives off the arbitrary variation, and require the remaining coefficient to vanish.

12 First variation versus second variation

The condition $\delta S = 0$ identifies a stationary path. To classify it, examine the quadratic term in

$$S[q+\varepsilon\eta] = S[q] + \varepsilon\,\delta S + \frac{1}{2}\varepsilon^2\,\delta^2 S + \cdots.$$

A positive second variation for every nonzero admissible η indicates a local minimum. An indefinite second variation indicates a saddle. Mechanics usually begins with the first variation because it already supplies the equations of motion.

13 A reliable derivation checklist

1. State the functional and the allowed class of paths.
2. Write $q_\varepsilon = q + \varepsilon\eta$.
3. Translate endpoint restrictions into conditions on η.
4. Differentiate with respect to ε at $\varepsilon = 0$.
5. Integrate by parts until derivatives no longer act on η.
6. Keep or justify every boundary term.
7. Use the arbitrariness of the interior variation.
8. Check the resulting equation against a familiar special case.

One-sentence summary

The Euler–Lagrange equation is the pointwise condition forced on a path when the first-order change of an integral functional vanishes for every allowed infinitesimal deformation of that path.